Fiscal Year:	FY 2023	**Task Last Updated:**	FY 12/23/2022
PI Name:	Beheshti, Afshin Ph.D.		
Project Title:	Circulating miRNAs Provides Systemic Host Response to Microgravity: Utilizing GeneLab Datasets to Identify Molecular Targets for Spaceflight Studies		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	PI has moved from NASA Ames Research Center to the University of Pittsburgh, where he serves as Director of the Center for Space Biomedicine. Former email: afshin.beheshti@nasa.gov		
Project Type:	Ground,NASA GeneLab	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	03/01/2019	**End Date:**	02/28/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/28/2023 per F. Hernandez/ARC (Ed., 4/25/22) NOTE: End date changed to 2/28/2022 per F. Hernandez/ARC (Ed., 5/5/21) NOTE: End date changed to 9/30/2021 per F. Hernandez/ARC (Ed., 10/20/20)		
Key Personnel Changes/Previous PI:	Update on 12/2/2020: Dr. Elge Cekanaviciute will no longer be getting funds since she has become a civil servant. That money will go back to Dr. Beheshti for research to use on the grant. Ed. note 11/6/2019: CoInvestigators have changed per approval of NASA Ames--Dr. Peter Grabham replaces original CoI Dr. John McDonald and Dr. Elge Cekanaviciute is added as CoI to assist Dr. Charles Vanderburg who was a previous Co-I and now is a consultant on the grant.		
COI Name (Institution):	Blaber, Elizabeth Ph.D. (Rensselaer Polytechnic Institute grant 80NSSC20K0307 (formerly NASA ARC employee)) Grabham, Peter Ph.D. (Columbia University grant 80NSSC19K1658)		
Grant/Contract No.:	Internal Project		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Biological risks associated with microgravity are a major concern for long-term space travel. Although determination of risk has been a focus for NASA biological research, data examining systemic (i.e., multi- or pan-tissue) responses to spaceflight are sparse. For this proposal, we utilized the NASA GeneLab database to analyze the transcriptomic data from 7 different rodent tissue datasets. Using our systems biology expertise, we were able to reveal for the first time the existence of potential "master regulators" coordinating systemic responses to microgravity. We identified p53, Transforming Growth Factor Beta 1 (TGFB1), and immune related pathways as the highly prevalent pan-tissue signaling pathways that are affected by microgravity responses. Key genes/drivers isolated by our unbiased systems biology analyses displayed a major overlap between tissue types and established that TGFB1 was the most inter-connected gene across all datasets. Finally, a hypothesis was generated indicating that an overall systematic microgravity response driven by TGFB1 is caused by a circulating microRNA (miRNA) signature. This miRNA signature, consisting of 13 significantly expressed miRNAs, was predicted from differential gene expression that causes miRNA-driven microgravity-related functional states. A subsequent impact on health using a theoretical "health risk score" was calculated. The genes and miRNAs identified from our analyses will be used for the ground based studies which will be proposed for this solicitation and will be investigated to develop potential countermeasures against microgravity response. This proposal exemplifies the utility of the GeneLab data repository to aid in the process of performing novel hypothesis-based spaceflight research aimed at elucidating the global impact of environmental stressors at multiple biological scales.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This miRNA signature can be a minimally invasive biomarker to monitor health risk during deep space missions. The antagomir countermeasures being developed can act as a preventive measure against space radiation and microgravity effects. Countermeasures can be applied throughout mission to potentially mitigate any space environment effects. The same techniques can be used in the clinic to monitor various diseases such as muscle degeneration risk in patients. miRNA signature can be used during yearly checkup as another diagnostic blood based test to monitor risk for these diseases. If miRNA signature starts to appear in patients, then can apply antagomir treatment to suppress active miRNAs that are associated with disease progression long before the manifestation of the disease will occur. We have shown through this proposal that antagomirs can potentially be a very promising countermeasure for mitigating spaceflight health risks.
	I) Summary of research originally proposed. Biological risks associated with microgravity are a major concern for long-term space travel. Although determination of risk has been a focus for NASA biological research, data examining systemic (i.e., multi- or pan-tissue) responses to spaceflight are sparse. For this proposal, we utilized the NASA GeneLab database to analyze the transcriptomic data from 7 different rodent tissue datasets. Using our systems biology expertise, we were able to reveal for the first time the existence of potential “master regulators” coordinating systemic responses to microgravity. We identified p53, TGFß1 and immune-related pathways as the highly prevalent pan-tissue signaling pathways that are affected by microgravity responses. Key genes/drivers isolated by our unbiased systems biology analyses displayed a major overlap between tissue types and established that TGFß1 was the most inter-connected gene across all datasets. Finally, a hypothesis was generated indicating that an overall systematic microgravity response driven by TGFß1 is caused by a circulating microRNA (miRNA) signature. This miRNA signature, consisting of 13 significantly expressed miRNAs, was predicted from differential gene expression that causes miRNA-driven microgravity-related functional states. A subsequent impact on health using a theoretical “health risk score” was calculated. The genes and miRNAs identified from our analyses will be used for the ground based studies which will be proposed for this solicitation and will be investigated to develop potential countermeasures against microgravity response. This proposal exemplifies the utility of the GeneLab data repository to aid in the process of performing novel hypothesis-based spaceflight research aimed at elucidating the global impact of environmental stressors at multiple biological scales. We hypothesize that microgravity-related changes in the transcriptome are caused by circulating miRNAs resulting in widespread alterations to TGFß1 expression. The administration of antagonists to the circulating miRNA signature associated to microgravity can be used as a countermeasure to TGFß1 dysregulation in microgravity, resulting in normalization of altered gene expression patterns. For this project we will confirm the presence of the miRNA signature using tissue samples from archived and ongoing rodent space research experiments to test our generated hypothesis. We will also perform in vitro experiments on various human cell lines by manipulating the miRNA expression to determine the cellular impact of the microgravity associated miRNA signature. Lastly, a novel miRNA countermeasure against microgravity will be developed and tested. The following specific aims will be used to achieve the goals of this proposal: SPECIFIC AIM 1: To confirm the existence of the hypothesized circulating miRNA signature that drives the dysregulation of TGFß1 microgravity response and determine the functional impact. The hypothesized circulating miRNA signature will be confirmed in archival tissue including serum and other tissue from rodents flown in space and future rodents that will be flown to the International Space Station (ISS). We will utilize droplet digital polymerase chain reaction (ddPCR) techniques to quantify the circulating miRNAs and TGFß1 expression in all samples. The validated miRNA signatures will then be inhibited utilizing established CRISPR/Cas9 techniques (CRISPR Therapeutics) and overexpressed (via viral techniques) in human cell lines to determine functional impact of the miRNA signature. SPECIFIC AIM 2: Develop countermeasure strategies against microgravity response based on antagonists to the microgravity associated circulating miRNA signature and TGFß1. Antagonists to the key circulating miRNA signature will allow for both mechanistic understanding and development of a novel countermeasure for microgravity health risk. Synthesized self-delivery miRNA antagonist (antagomirs) will be tested with established in vitro and in vivo simulated microgravity techniques. In addition, we will use established TGFß1 reagents for both in vitro and in vivo experiments to promote TGFß1 signaling and to also develop additional countermeasures for the hypothesized master regulator gene being impacted due to spaceflight. This will allow multiple countermeasures to be developed and determination of the optimal technique to be assessed. Quantification of the miRNA signature and TGFß1 expression will be assessed using ddPCR. Systems biology techniques (i.e., pathway analysis and in silico predictions) will be further used to determine the overall global impact on molecular pathways. Lastly, any possible toxicity due to the countermeasures will be

Task Progress:

determined in the in vivo models being utilized for this aim.

II) Summary of accomplishments.

We have fully characterized the miRNA spaceflight signature and its role with different models that were exposed to the space environment. For experimental validation of the role of spaceflight-associated miRNAs in rodent models, we have investigated their changes in response to individual spaceflight stressors: microgravity using hindlimb unloading (HU), as well as ionizing radiation (IR) to simulate galactic cosmic rays using their chief components: protons and/or high mass and charge 56Fe particles. We have exposed male 16-week old mice to HU, IR, or their combination, and measured miRNAs in their serum at 1 and 11 days post exposure using ddPCR.

To examine the contribution of reduced gravity to the miRNA signature at higher resolution, a partial weight bearing (PWB) Wistar male rat model was used to simulate varying degrees of gravitational loading. We observed that decreasing amounts of gravitational loading increased the amount of miRNAs of interest, suggesting a gravity-dependent dose response. These results also validated a reduced-gravity-specific miRNA signature in both mice and rats and separated the miRNAs associated with reduced gravity from those that associated with exposure to simulated deep space radiation in mice. The only observed exception was miR-217, which did not change. A similar miRNA dose response was observed based on correlation with muscle mass. The miRNAs of interest increased with reduced muscle mass (which is associated with lower simulated gravity both in this rat model and in astronauts). In addition, we have data that shows the impact of these miRNAs at low-Earth orbit (LEO) from serum in mice that were flown to the ISS for 30 days. This data is available in the related publication and will not be shown here to save space.

To understand the specificity and variation of spaceflight-relevant miRNA levels within the global miRNA population, miRNA-seq was performed on tissue samples from 16 week old female C57BL/6J mice after exposure to simulated spaceflight conditions: normal loading (NL) or hindlimb unloading (HU) for two weeks prior to irradiation with 5 Gy gamma, 1 Gy simulated solar particle event (simSPE), 0.5 Gy simulated simplified galactic cosmic rays (SimGCRSim), or sham control. SimGCRSim consists of 5 ions: protons at 1000MeV, 28Si at 600 MeV/n, 4He at 250 MeV/n, 16O at 350 MeV/n, 56Fe at 600 MeV/n, and protons at 250 MeV. This dose of radiation is equivalent to what an astronaut is predicted to receive in deep space during a Mars mission, though it is modeled as a single exposure over 25 minutes instead of the actual chronic exposures over 1.5 years. SimSPE consists of protons from 50 MeV to 150 MeV. 24 hours after irradiation, the mice were sacrificed and miRNA-seq was performed on tissue from plasma, heart, liver, and soleus muscle. From the miRNA-seq data we were able to uncover a common miRNA-seq profile with overlapping existing miRNA family members with the original miRNA spaceflight signature. In addition, the miRNA spaceflight signature also has appeared in various tissues for these mice. Lastly, based on the global pathway analysis on the miRNA-seq data, we observed global regulation of downstream targets that overlaps with the miRNA spaceflight signature. This work was done with Dr. Elizabeth Blaber (Rensselaer Polytechnic Institute, grant #80NSSC20K0307).

In collaboration with Dr. Christopher Mason (Weill Medical College of Cornell University), we also were able to confirm the presence of this spaceflight miRNA signature in humans from data available on the NASA Twins Study. In addition, in collaboration with Dr. Sylvain Costes (NASA Ames Research Center), we also confirmed the presence of this miRNA signature with human peripheral blood mononuclear cells (PBMCs) that were irradiated with 0.3Gy 600MeV/n 56Fe particles. From the NASA Twins Study data, we analyzed miRNA-sequence data, single-cell RNA sequencing (scRNA-seq), and single-cell sequencing Assay for Transposase-Accessible Chromatin (scATAC-seq) on the blood and different cell types from the blood. The miRNA downstream targets were determined in different cell types from the scRNA-seq and scATAC-seq data. The figure for this data is available in the publication in Cell Reports (Ed. Note: see References below and the NASA Task Book Cumulative Bibliography for this project).

Finally, we have developed and utilized a mature human 3D microvascular tissue model seeded with human umbilical vein endothelial cells (HUVECs) to examine the molecular, cellular, and physiological effects of candidate miRNAs on cardiovascular damage. This work was done with Dr. Peter Grabham (Columbia University, grant #80NSSC19K1658). Several of the candidate miRNAs showed marked increases at 48 hours after 0.5 Gy irradiation with simulated 5-ion simplified galactic cosmic rays (SimGCRSim) irradiation compared to controls, though the increases were not significant, possibly due to low sample numbers. Note that this dose of irradiation is equivalent to what an astronaut will receive in a deep space. Three of the increased miRNAs (miR-125, miR-16, and let-7a) have previously been predicted to be associated with space radiation induced cardiovascular damage, and were consequently chosen to be inhibited using antagomirs. In addition, we observed irradiation-mediated loss of vascularization as quantified by a decreased green fluorescent protein (GFP) signal from GFP stained HUVECs, indicating that the vessels had shrunk or collapsed due to galactic cosmic ray GCR irradiation. The impact of the space radiation was rescued by applying three antagomirs to inhibit miR-125, miR-16, and let-7a at 24 hours prior IR and 48 hours and 72 hours post IR. All this data is available in our Cell Reports publication which is referenced below.

We also have done additional experiments that show the key biological changes associated with the antagomir treatment for the 3 miRNAs. This work has been submitted for peer review and is currently under review in Nature Communications. This further showed the key biology that has been rescued by the antagomir countermeasure treatment. We showed that the following key pathways were restored back to control levels: inflammation, immune, key mitochondrial pathways, and improved efficiency of DNA double strand break (DSB) repair.

III) Summary of risks or obstacles, plus mitigation strategies.

COVID-19 shut down NASA Ames Research Center and Broad Institute in April 2020 (and the institutions did not fully open until 2021). The shutdown has provided some obstacles in getting work done. Despite this obstacle, we have still been able to successfully publish a couple of papers related to this grant due to our earlier findings. We were able to get extensions for this grant that allowed us to get the rest of the work needed to get done.

IV) Future directions

Dr. Beheshti is planning on continuing both the functional and countermeasure experiments with the other CoInvestigators (CoIs) on the grant (Drs. Elizabeth Blaber and Peter Grabham) for future projects and grant submissions. This can lead to novel and new countermeasures to allow for safe space travel.

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